

MCQS

PHY301 midterm preparation file

ZB

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

In the name of God , Most
Gracious, Most merciful



PHY301

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1. What is the unit of force in the SI system?

- A) Newton**
- B) Joule**
- C) Watt**
- D) Pascal**

Answer: A) Newton

2. The velocity of a particle in uniform circular motion is:

- A) Constant**
- B) Changing in magnitude**
- C) Changing in direction only**
- D) Changing in both magnitude and direction**

Answer: C) Changing in direction only

3. The work done by a force is given by:

- A) Force $\times$ Distance**
- B) Force $\times$ Time**
- C) Power $\times$ Time**
- D) Power $\times$ Distance**

Answer: A) Force $\times$ Distance

5. The center of mass of a system of particles moves as if it were a single particle of mass equal to the total mass of the system under the action of:

- A) The net external force
- B) The net internal force
- C) All the forces
- D) The gravitational force

Answer: A) The net external force

Thermodynamics:

6. The first law of thermodynamics is a statement of the conservation of:

- A) Energy
- B) Mass
- C) Momentum
- D) Entropy

Answer: A) Energy

8. The entropy of an ideal gas increases when:

- A) Temperature increases at constant volume**
 - B) Pressure increases at constant temperature**
 - C) Volume increases at constant temperature**
 - D) Temperature decreases at constant pressure**
- Answer: C) Volume increases at constant temperature**

9. In an adiabatic process, the heat exchanged is:

- A) Zero**
 - B) Constant**
 - C) Equal to the work done**
 - D) Equal to the change in internal energy**
- Answer: A) Zero**

10. Which of the following is a state function?

- A) Heat**
 - B) Work**
 - C) Internal energy**
 - D) Power**
- Answer: C) Internal energy**

11. Coulomb's law gives the force between two point charges. The magnitude of this force is inversely proportional to:

- A) The distance between the charges
- B) The square of the distance between the charges
- C) The sum of the charges
- D) The difference of the charges

Answer: B) The square of the distance between the charges

12. The magnetic field at the center of a current-carrying loop is proportional to:

- A) The radius of the loop
- B) The current
- C) The square of the radius of the loop
- D) The distance from the center of the loop

Answer: B) The current

13. In an electric field, the electric potential energy of a charge is zero at:

- A) Infinity
- B) The surface of a conductor
- C) The center of the electric field
- D) The midpoint between two charges

Answer: A) Infinity

14. The unit of electric potential is:

- A) Coulomb**
- B) Newton**
- C) Volt**
- D) Ampere**

Answer: C) Volt

15. The force on a charged particle moving in a magnetic field is maximum when the angle between velocity and magnetic field is:

- A) 0°**
- B) 45°**
- C) 90°**
- D) 180°**

Answer: C) 90°

Optics:

16. The focal length of a concave lens is:

- A) Positive**
- B) Negative**
- C) Zero**
- D) Infinite**

Answer: B) Negative

17. The speed of light in a medium depends on the:

- A) Frequency of light
- B) Wavelength of light
- C) Refractive index of the medium
- D) Angle of incidence

Answer: C) Refractive index of the medium

18. The image formed by a concave mirror when the object is beyond the focal point is:

- A) Virtual, erect, and diminished
- B) Virtual, inverted, and diminished
- C) Real, inverted, and diminished
- D) Real, inverted, and magnified

Answer: D) Real, inverted, and magnified

19. The refractive index of a medium is defined as the ratio of:

- A) Speed of light in vacuum to the speed of light in the medium
- B) Speed of light in the medium to the speed of light in vacuum
- C) Wavelength in vacuum to the wavelength in the medium
- D) Wavelength in the medium to the wavelength in vacuum

Answer: A) Speed of light in vacuum to the speed of light in the medium

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20. The angular resolution of a telescope depends on:

- A) The focal length of the lens**
 - B) The diameter of the lens**
 - C) The wavelength of light**
 - D) The distance of the object**
- Answer: B) The diameter of the lens**

Waves and Sound:

21. The frequency of a wave is:

- A) The distance traveled by the wave per unit time**
 - B) The time taken for one complete cycle of the wave**
 - C) The number of cycles per unit time**
 - D) The distance between two consecutive crests**
- Answer: C) The number of cycles per unit time**

22. When the frequency of a sound wave increases, the wavelength:

- A) Increases**
 - B) Decreases**
 - C) Remains the same**
 - D) Is irrelevant to frequency**
- Answer: B) Decreases**

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23. The Doppler effect is observed when there is a change in the frequency of a wave due to:

- A) The movement of the wave source
- B) The movement of the observer
- C) Both the movement of the wave source and observer
- D) The temperature of the medium

Answer: C) Both the movement of the wave source and observer

24. The speed of a sound wave in air depends on:

- A) The temperature of the air
- B) The density of the air
- C) The pressure of the air
- D) The frequency of the wave

Answer: A) The temperature of the air

25. The intensity of a sound wave is proportional to:

- A) The square of the amplitude
- B) The square root of the amplitude
- C) The amplitude
- D) The inverse of the amplitude

Answer: A) The square of the amplitude

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30. The gravitational force between two objects decreases as the distance between them:

- A) Decreases
- B) Increases
- C) Remains the same
- D) Doubles Answer: B) Increases

Thermodynamics (continued):

31. In an isothermal process, the temperature of the system:

- A) Increases
- B) Decreases
- C) Remains constant
- D) Varies according to pressure Answer: C) Remains constant

32. For an ideal gas, the relationship between pressure , volume , and temperature is given by:

- A) Boyle's law
- B) Charles' law
- C) The ideal gas law
- D) Gay-Lussac's law Answer: C) The ideal gas law

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33. During an isobaric process, the pressure of the system:

- A) Remains constant
- B) Increases
- C) Decreases
- D) Increases and decreases alternatively Answer: A) Remains constant

34. In the Carnot cycle, the maximum efficiency occurs when:

- A) The temperature of the hot reservoir is very high
- B) The temperature of the cold reservoir is very low
- C) The difference between the hot and cold reservoir is maximized
- D) The pressure difference is maximized Answer: C) The difference between the hot and cold reservoir is maximized

35. In a thermodynamic system, the change in internal energy is equal to:

- A) Heat + Work
- B) Heat - Work
- C) Heat
- D) Work Answer: B) Heat - Work

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36. The direction of the magnetic field at any point is given by:

- A) The magnetic force on a moving charge**
- B) The current in the wire**
- C) The right-hand rule**
- D) The direction of the current in a wire** Answer: C) The right-hand rule

37. The force on a current-carrying wire placed in a magnetic field is:

- A) Proportional to the current, length of the wire, and magnetic field**
- B) Independent of the current**
- C) Independent of the magnetic field**
- D) Inversely proportional to the length of the wire** Answer: A) Proportional to the current, length of the wire, and magnetic field

38. The total electric flux through a closed surface depends on:

- A) The surface area**
- B) The electric field**
- C) The charge enclosed within the surface**
- D) The potential difference across the surface** Answer: C) The charge enclosed within the surface

39. The induced electromotive force (emf) in a coil is given by:

- A) Faraday's law
 - B) Lenz's law
 - C) Ampere's law
 - D) Coulomb's law
- Answer: A) Faraday's law

40. The electric field inside a conductor in electrostatic equilibrium is:

- A) Zero
 - B) Constant
 - C) Proportional to the charge on the conductor
 - D) Perpendicular to the surface of the conductor
- Answer: A) Zero

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WhatsApp:

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Passing criteria of this course

Letter Grade	Grade Points	Equivalent Percentage
A+	4.00	90-100
A	4.00	85-89
A-	3.66 - 3.99	80-84
B+	3.33 - 3.65	75-79
B	3.00 - 3.32	71-74
B-	2.66 - 2.99	68-70
C	2.00 - 2.65	61-67
D	1.00 - 1.99	50-60
F	0.00	00-49

Passing Criteria for a Course:

The passing criteria is defined in such a way that it ensures the student shall be consistent in his studies throughout the semester.

Therefore, for passing a course/subject, student shall fulfill the following:

- Secure minimum **20%** score in Formative Assessments/Mid terms
- Secure minimum **20%** score in Final Term Examinations.
- Secure at least **40%** marks in aggregate while fulfilling the above requirements

Course Selection and Credit Hours

Q: How do I select courses when the course selection is open?

A: Follow these steps:

- 1. Check the credit hours allowed by your university (e.g., 21, 18, or 15 credit hours).**
- 2. Each subject typically has 3 credit hours.**
- 3. Divide the total allowed credit hours by 3 to determine how many subjects you can select.**

Examples:

21 credit hours = 7 subjects

15 credit hours = 5 subjects

Remember, the university may allow different credit hours for each student, so check your specific allowance.

By following these steps, you can make informed decisions during course selection and manage your credit hours effectively.

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Advice

Hmaesha relax ho ke parha kryn panic mat hua kryn apky parents ky bohat khawab hoty hn wo pura krny ki puri koshish kea kryn kbi b ksi pe depend na kea kryn apko bs ak insan success kr saqta ha or wo insan ap khud hn apky elawa koi nahe ap bs koshish kryn Allah pak pe strong yakeen rakha kryn or mehnat krke sb Allah ke hawaly kr dea kryn everything is possible be brave be strong stay blessed

Hum insan hamesha moat se darty or bhagty hn or moat se bachny ki koshish krty hn jabke moat ny ana hi ana ha hum

moat se nahe bach saqty humy jahanam se bachny ki koshish
krni chaheay hum jahanam se bach saqty hn
Insan ko 3 chezo se dar lagta ha Moat Risk/Dolat Ezat Shohrat
Fame

Moat

**Humy pta hona chaheay Moat tab ani ha jab Allah
chahy us se pehly puri dunya bi ak taraf ho ke apko
marna chahy to apka kuch nahe beggar saqti**

Risk/Dolat

**Risk Dolat sb Allah pak ke hath m ha puri dunya ak
taraf b ho jay na apsy apka risk cheen saqti ha na de
saqti ha Ye sirf ALLAH pak ky hath m ha**

Izat Zilat shohrat Fame

**Izat Zilat Sirf Allah de saqty hn puri dunya mel ke b
apko zra brabr b damage nahe kr saqti Agr ALLAH**

**Apko izat dena chahay to puri dunya ak taraf ho ke
b apko 1% b nuksan nahe pohancha saqti**

Phr Dar ks bat ka ??????????????????

**Be brave be strong Just put your trust To ALLAH
Hamesha Confident or himat se raho kbi ksi k samne
mat jhuko puri dunya mel k b apka kuch b nahe
beggar saqti**

ZB
MY REQUEST FROM ALL OF YOU

ZB Request from All of you My family

Mjy ap sbki help or support ki zarort ha or wo ye k hum sb Mel kr Allah ka Quran ki Urdu translation logo tak pohanchy or Quran ko samjna asaaan kryn dosro k leay sb tak Allah ka Quran pohanchy it's my campaign hum log 70+ age k ho jaty hn phr b hmy namaz tak ki translation nahe pta hoti k hum Allah pak se

Kya Dua kr rhy hn so hum youngster's ko Mel kr puri takat or energy ke sath Allah ka Quran spread krna ha

Hum puri Koshish kr rhy hn k hum sb Mel k Quran ki translation or most important topics ko maximum share kry with translation hmara maksad Quran ko spread krna ha be a part of us

m apni pocket se ye sb kr rha ho Allah ka Quran spread krny k leay apne or mene sbny Marna ha ak din to q na Allah k leay Kuch kryn apni energies apni power ko bajay negative use krny k Allah k leay invest kryn khud ko be a part of us

Agr m 23 ki age m apny sare sources use kr rha ho pocket se heavy amounts give away kr rha ho Quran spread krny k leay to ap just share to kr saqty hn itna e krdyn

Contact ZB [+92 300 5427579](tel:+923005427579)

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Agr ap mujsy contact na kr sky too b ap jo b hn jaha bi hn waha Allah ka quran pheelay zada se zada with urdu translation jitni himat ha utna share kryn

..... **MY Family**

**ust Relax and focus! Exams are not difficult. Put in your
100% effort and trust in Allah.**

..... **AL-Quran**

**"And indeed, with hardship comes ease." (Quran 94:5)
Remember, I'm here to support you! Stay brave and strong!"**

May Allah bless you